A Symplectic-Geometric Perspective on Functional Theories

Julia Liebert

Princeton University

In collaboration with:

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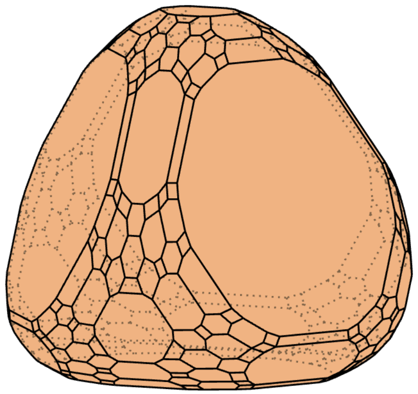

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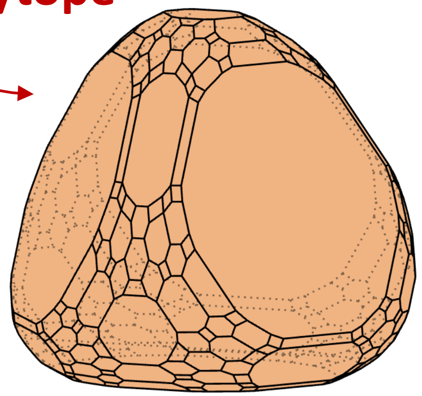
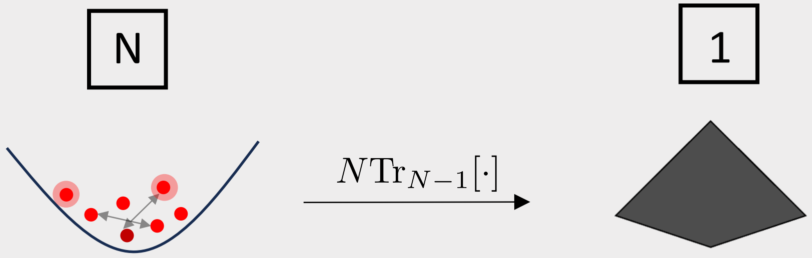
Moment Polytope

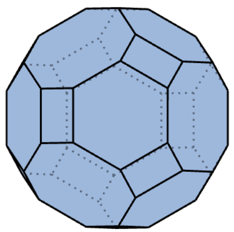


N-representability Problem

(Quantum marginal problem)

"Which reduced density matrices are compatible with an N-fermion quantum state?"





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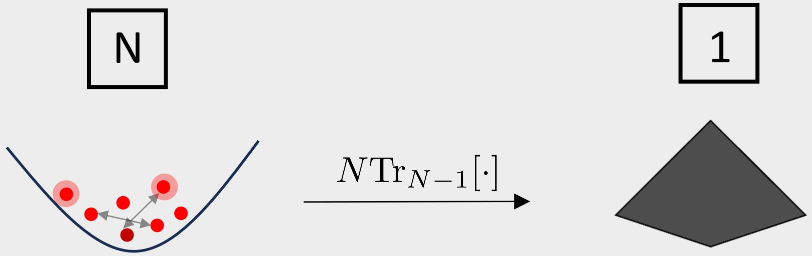
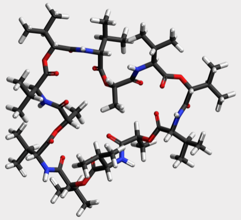
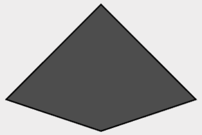
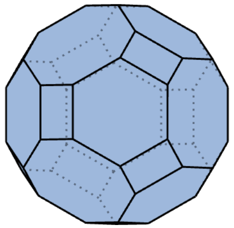


Functional Theories



Solution to ground state problem?

$\mathcal{F}:$  $\mapsto \mathbb{R}$



N-representability Problem

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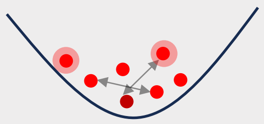
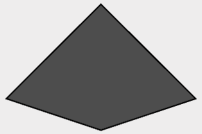
"Which reduced density matrices are compatible with an N-fermion quantum state?"

N

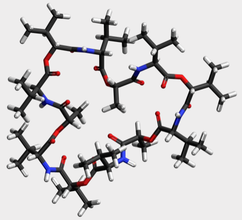
1



$$\xrightarrow{N\mathrm{Tr}_{N-1}[\cdot]}$$



Functional Theories



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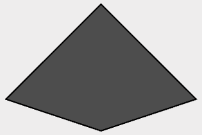
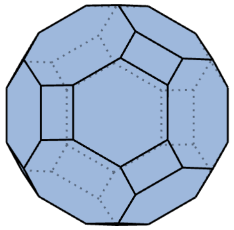
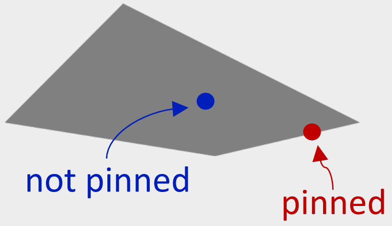
$\mathcal{F}$:  $\mapsto \mathbb{R}$



Relevance of (symmetry-adapted) Generalized Pauli Constraints

"Do natural occupation number vectors lie on the boundary?"



$\Rightarrow$ Reassessment of pinning under spin symmetries

[C. Schilling, M. Altunbulak, S. Knecht, A. Lopes, J.D. Whitfield, M. Christandl, D. Gross, M. Reiher, Phys. Rev. A **97**, 052503, (2018),
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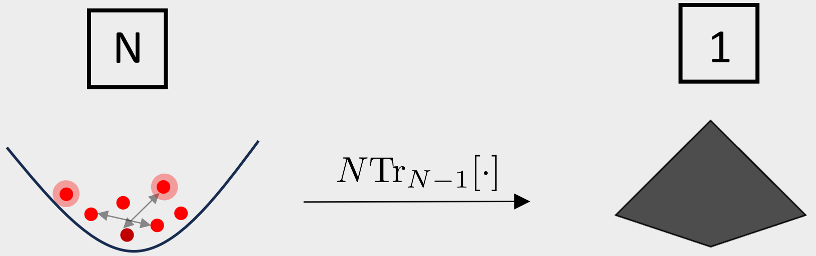

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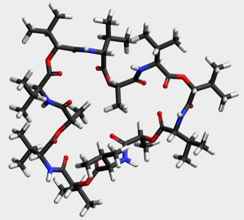

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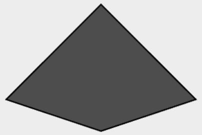
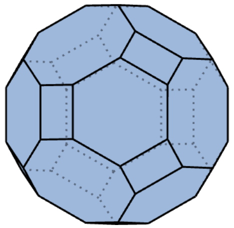
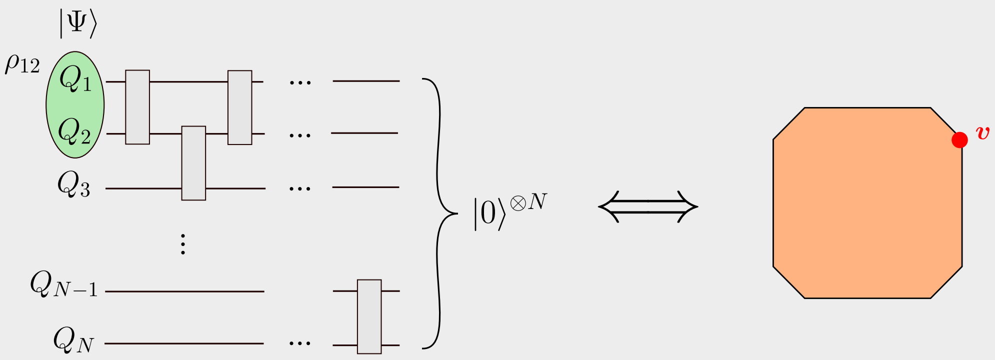
$\mathcal{F}$:  $\mapsto \mathbb{R}$



Disentangling Algorithm



[S. Jevtic, D. Jennings, and T. Rudolph, Phys. Rev. Lett. **108**, 110403 (2012),
L. Cigna, F. Castillo, J.-P. Labbé, J. Liebert, C. Schilling, forthcoming (2025)]

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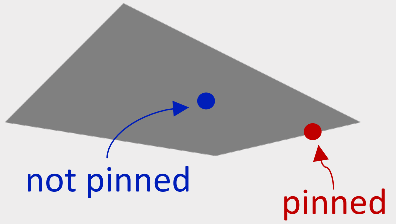

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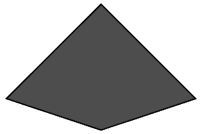
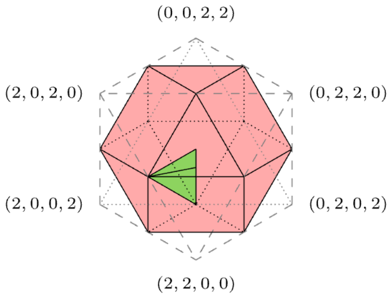
Contents

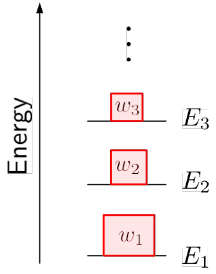
I) Functional Theories from Moment Maps

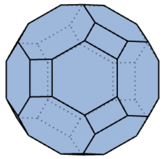
$\mathcal{F}:$  $\mapsto \mathbb{R}$

II) Spin Symmetries



III) Excited States





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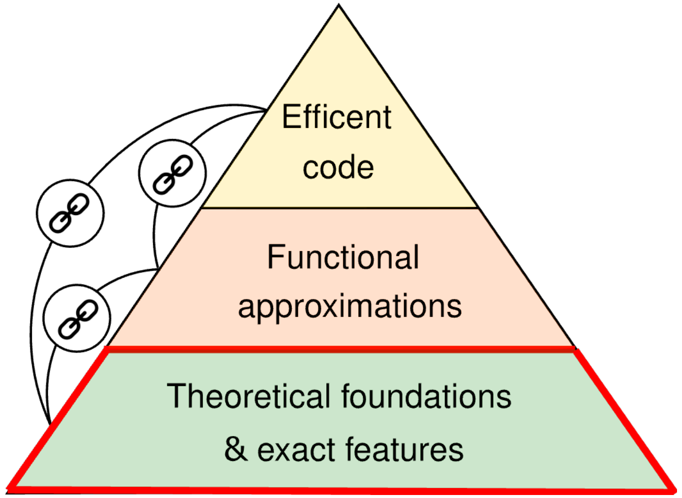


Key Concepts

<u>*N*-fermion Hilbert space:</u> $\mathcal{H}_N = \wedge^N \mathcal{H}_1 \leq \mathcal{H}_1^{\otimes N}$

<u>Idea:</u> Start from symmetry group and moment map to define both natural variable of functional and domain

Key Concepts

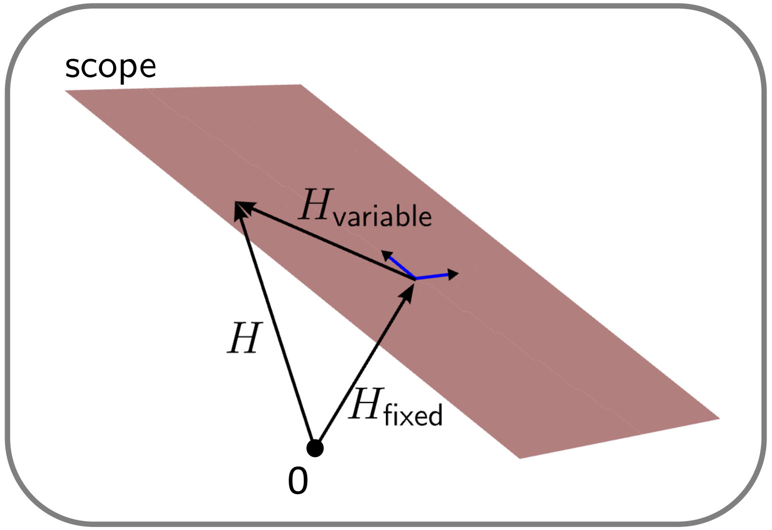
N-fermion Hilbert space: $\mathcal{H}_N = \wedge^N \mathcal{H}_1 \leq \mathcal{H}_1^{\otimes N}$

Idea: Start from symmetry group and moment map to define both natural variable of functional and domain

1) Scope of a functional theory:

$\Rightarrow$ Split Hamiltonian into 'variable' and 'fixed' part:

$$\boxed{H = H_{\text{variable}} + H_{\text{fixed}}}$$

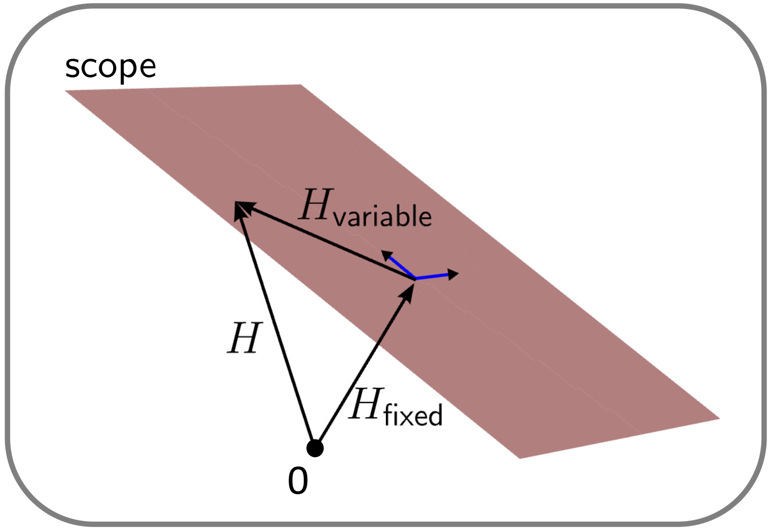


<u>Idea:</u> Start from symmetry group and moment map to define both natural variable of functional and domain

- $H_{\text{variable}} \in i\mathfrak{g}$...real linear vector space of Hermitian matrices
- $\mathfrak{g}^*$...dual of $\mathfrak{g}$

Dual pairing:

$\langle \rho, X \rangle \quad \rho \in \mathfrak{g}^*, X \in i\mathfrak{g}$



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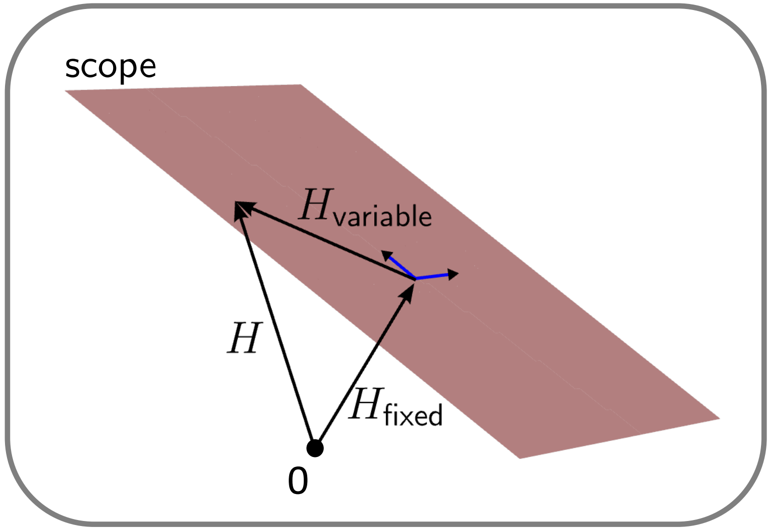
2) Moment map:

$$\mu : \mathbb{P}(\mathcal{H}_N) \to \mathfrak{g}^*$$

$$\langle \mu(\Psi), X \rangle := \frac{\langle \Psi, X\Psi \rangle}{\langle \Psi, \Psi \rangle} \quad \forall [\Psi] \in \mathbb{P}(\mathcal{H}_N), X \in i\mathfrak{g}$$



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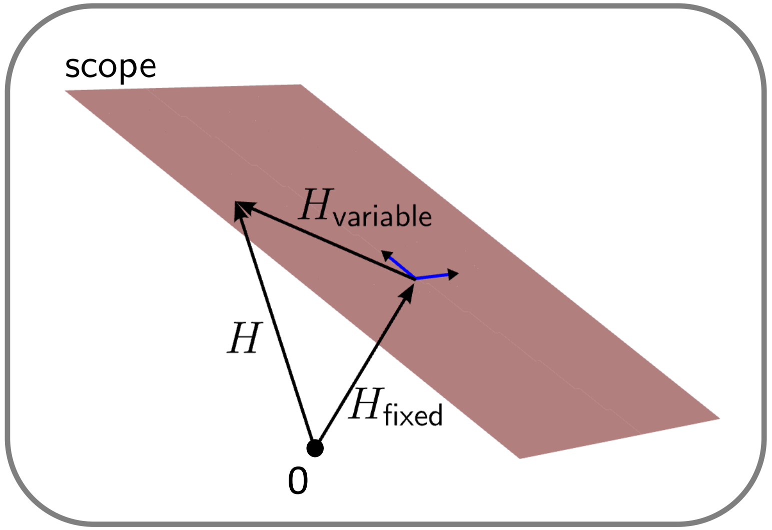
natural variable



<u>Idea:</u> Start from symmetry group and moment map to define both natural variable of functional and domain

<u>3) Constrained search:</u>

- $\mathcal{D} := \mu(\mathbb{P}(\mathcal{H}_N))$

Ground state energy

$$
\begin{aligned}
E_1(X) &= \inf_{[\Psi]\in\mathbb{P}(\mathcal{H}_N)} \frac{\langle\Psi,(X+H_{\mathrm{fixed}})\Psi\rangle}{\langle\Psi,\Psi\rangle} \\
&= \inf_{\rho\in\mathcal{D}}\left\{\inf_{\Psi\in\mu^{-1}(\rho)}\left[\langle\mu(\Psi),X\rangle + \frac{\langle\Psi,H_{\mathrm{fixed}}\Psi\rangle}{\langle\Psi,\Psi\rangle}\right]\right\} \\
&= \inf_{\rho\in\mathcal{D}}\left[\langle\rho,X\rangle + \mathcal{F}(\rho)\right]
\end{aligned}
$$

<u>Universal functional:</u>

$$
\boxed{\mathcal{F}(\rho) := \inf_{\Psi\in\mu^{-1}(\rho)} \frac{\langle\Psi,H_{\mathrm{fixed}}\Psi\rangle}{\langle\Psi,\Psi\rangle}}
$$

Density Functional Theory (DFT)

- $\mathcal{H}_1 \cong \mathbb{C}^d$
- Group: $G = T \cong U(1)^d$
- $i\mathfrak{g}$: local external potentials

Moment map:

$$\langle \mu(\Psi), v \rangle = \sum_{i=1}^{d} v_i n_i$$

local external potential

lattice site occupations

In the continuum setting:

- $\rho(\mathbf{r}) = N \int \mathrm{d}\mathbf{r}_2 ... \mathrm{d}\mathbf{r}_N |\Psi(\mathbf{r}, \mathbf{r}_2, ..., \mathbf{r}_N)|^2$
- $\mathcal{D} = \mu(\mathbb{P}(\mathcal{H}_N))$: set of N-representable densities

[E.H. Lieb, Int. J. Quantum Chem. **24**, 243 (1983),
G. Rosensteel, Ts. Dankova, J. Phys. A **31**, 8933-8940 (1998)]

Reduced Density Matrix Functional Theory (RDMFT)

- $\mathcal{H}_1 \cong \mathbb{C}^d \otimes \mathbb{C}^2 \quad \Rightarrow \quad G = U(2d)\,, \mathfrak{g} = \mathfrak{u}(2d)$

Moment map:

$$\langle \mu(\Psi), h \rangle = \mathrm{Tr}[h\gamma]$$

One-body reduced density matrix (1RDM):

$$\boxed{\gamma = N\mathrm{Tr}_{N-1}[\Gamma] = \sum_\alpha \lambda_\alpha |\alpha\rangle\langle\alpha|}$$

natural occupation numbers

natural orbitals

Atiyah/Guillemin-Sternberg:

$$\Sigma^{(p)\downarrow} := \mathrm{img}(\mu(\mathbb{P}(\mathcal{H}_N)) \cap \mathfrak{t}_+^*$$

is a convex polytope 

[M.F. Atiyah, Bull. London Math. Soc. **14**, 1-15 (1982),
V. Guillemin, S. Sternberg, Invent. Math. **67**, 491-513 (1982)]

Consequences for Functional Theories

Moment map:

- Choice of "minimal" natural variable
- Moment polytopes (images of moment map) provide solution to N-representability problem
- Equivariance of moment map: $\mu(g \cdot \Psi) = \mathrm{Ad}_g^* \mu(\Psi)\,,\ \ g \in G$

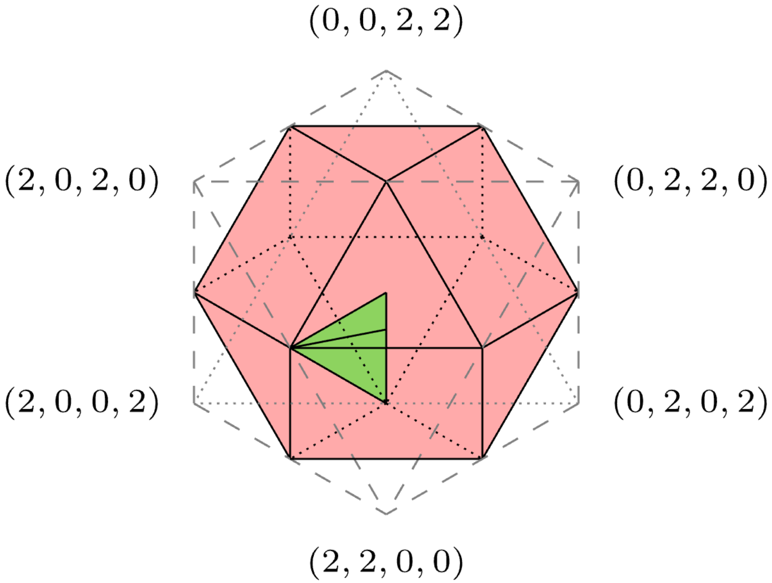
$\Longrightarrow$ Reduced variable transforms consistently with symmetry

$\Longrightarrow$ Symmetry constraints on exact functional

Alternative perspective on constrained search:

- Fibers $\mu^{-1}(\rho)$ $\Longrightarrow$ information missing in the reduced variable

II) Spin Symmetries



Foundations

- Spin-1/2 fermions: $\mathcal{H}_1 = \mathcal{H}_1^{(l)} \otimes \mathcal{H}_1^{(s)} \cong \mathbb{C}^d \otimes \mathbb{C}^2$

Orbital degrees of freedom

Spin degrees of freedom

<u>Relevant symmetry groups:</u> $\mathcal{S}_N \times U(2d) \supset \mathcal{S}_N \times U(d) \times U(2)$

<u>N-fermion Hilbert space:</u> $\mathcal{H}_N := \wedge^N \mathcal{H}_1 \cong \bigoplus_{\lambda \vdash N} V_{U(d)}^{\lambda^{\mathrm{T}}} \otimes V_{U(2)}^{\lambda}\,, \qquad \lambda_1 \leq d\,,\ \lambda_1^{\mathrm{T}} \leq 2$

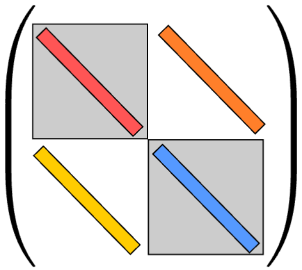
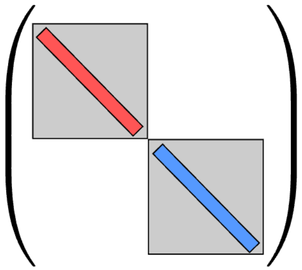
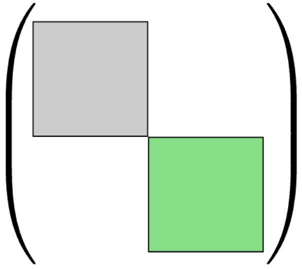
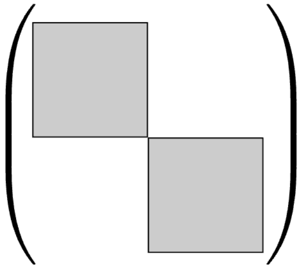
”Spin-adapted configuration state functions”

$\Longrightarrow$ Construct irreps of U(d) to solve the spin-adapted N-representability problem

Spin Symmetries

$$\mathcal{H}_N = \bigoplus_{S,M} \mathcal{H}_N^{(S,M)}$$

SU(2) invariance: $[h, S_i] = 0 \ \ \forall i \in \{x, y, z\} \quad \Rightarrow \quad [h, \boldsymbol{S}^2] = [H, S_z] = 0$

	$[h, \boldsymbol{S}^2] = 0$	$[h, \boldsymbol{S}^2] = [h, S_z] = 0$	$[h, S_z] = 0$	$[h, S_i] = 0 \ \forall i = x, y, z$
h				
ρ	(γ_l, γ_s)	$\gamma_l, (\gamma_s$ fixed)	$(\gamma_\uparrow, \gamma_\downarrow)$	γ_l

Orbital 1RDM: $\gamma_l := \mathrm{Tr}_{\mathcal{H}_1^{(s)}}[\gamma] = \gamma_\uparrow + \gamma_\downarrow$

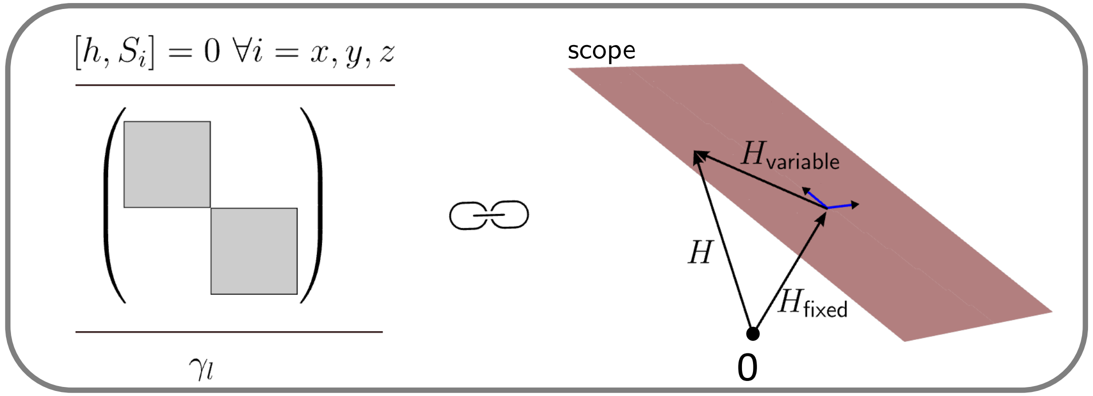
Spin 1RDM: $\gamma_s = \mathrm{Tr}_{\mathcal{H}_1^{(l)}}[\gamma]\,, \quad \mathrm{Tr}[\gamma_s] = 2M$

SU(2)-adapted RDMFT

$\mathcal{H}_1 \cong \mathbb{C}^d \otimes \mathbb{C}^2$

Scope: $\{H_{\text{variable}}\} \leftrightarrow i\mathfrak{u}(d) \quad h_l = \tilde{h}_l \otimes \mathbb{1}$

$H_{\text{fixed}} = W$ (interaction)



Moment map: $G = U(d)$

$$\mu : \mathbb{P}(\mathcal{H}_N^{(S,M)}) \to \mathfrak{u}^*(d)$$

$$\langle \mu(\Psi), h_l \rangle = \langle \gamma_l, h_l \rangle \,, \quad \gamma_l = \gamma_\uparrow + \gamma_\downarrow$$

$\Rightarrow$ Pure state (N, S, M)-representable orbital 1RDMs: $\mathcal{L}^1_{N,S,M} = \mu(\mathbb{P}(\mathcal{H}_N^{(S,M)}))$

described by spin-adapted generalized Pauli constraints (GPCs)

Constrained search:

$$\boxed{E_1^{(S,M)}(h_l) = \min_{\gamma_l \in \mathcal{L}^1_{N,S,M}} \left[\langle h_l, \gamma_l \rangle + \mathcal{F}^{(S,M)}(\gamma_l) \right]}$$

$\Rightarrow$ **Both** functional and domain depend on quantum numbers!

Spin-adapted N-representability problem

<u>Pure</u> state problem: spin-adapted GPCs

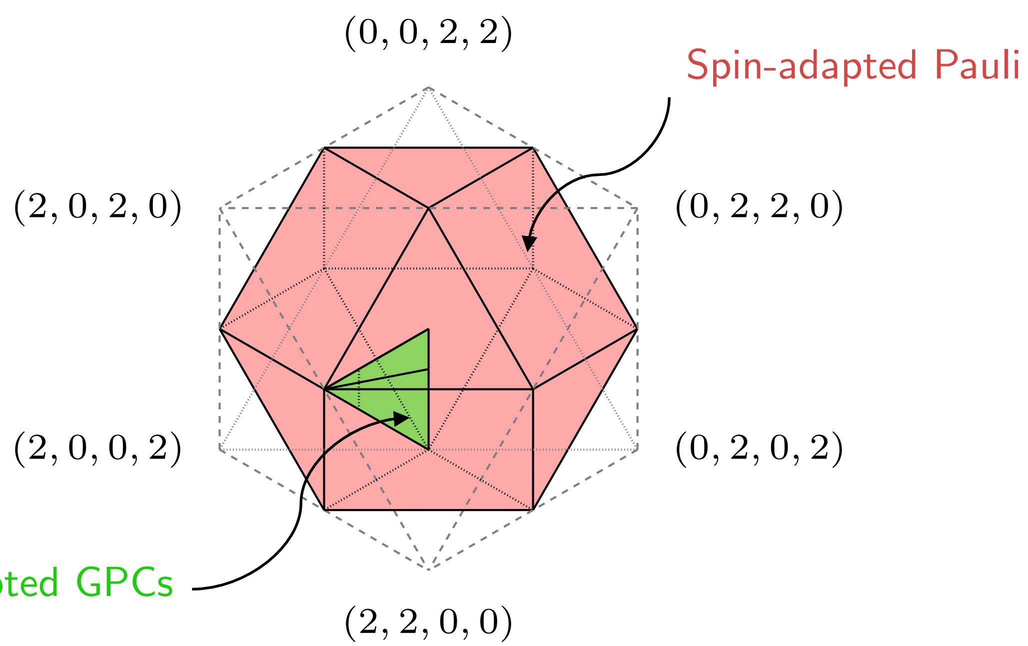
highly complicated!

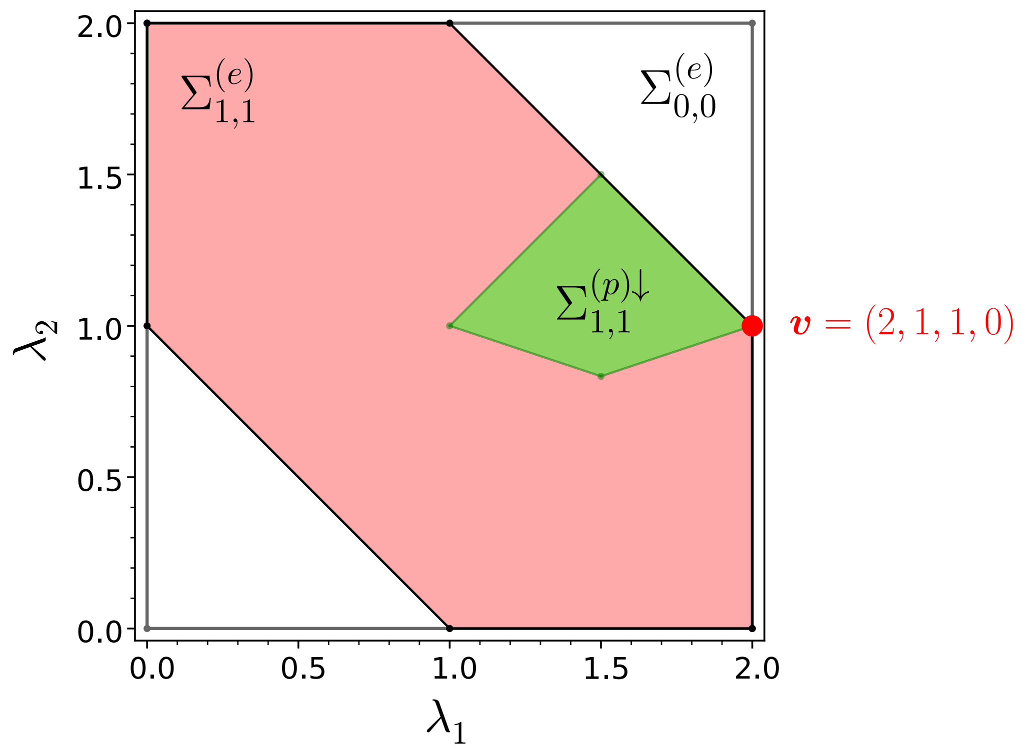
⟺

<u>Mixed</u> state problem: spin-adapted Pauli constraints

effectively (N,S,M) independent

Example: $S = M = 1, N = 4$





[M. Altunbulak, A. Klyachko, Commun. Math. Phys. **282**, 287 (2008),
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Pure state problem: spin-adapted GPCs

highly complicated!

$\Longleftrightarrow$

Mixed state problem: spin-adapted Pauli constraints

effectively (N,S,M) independent

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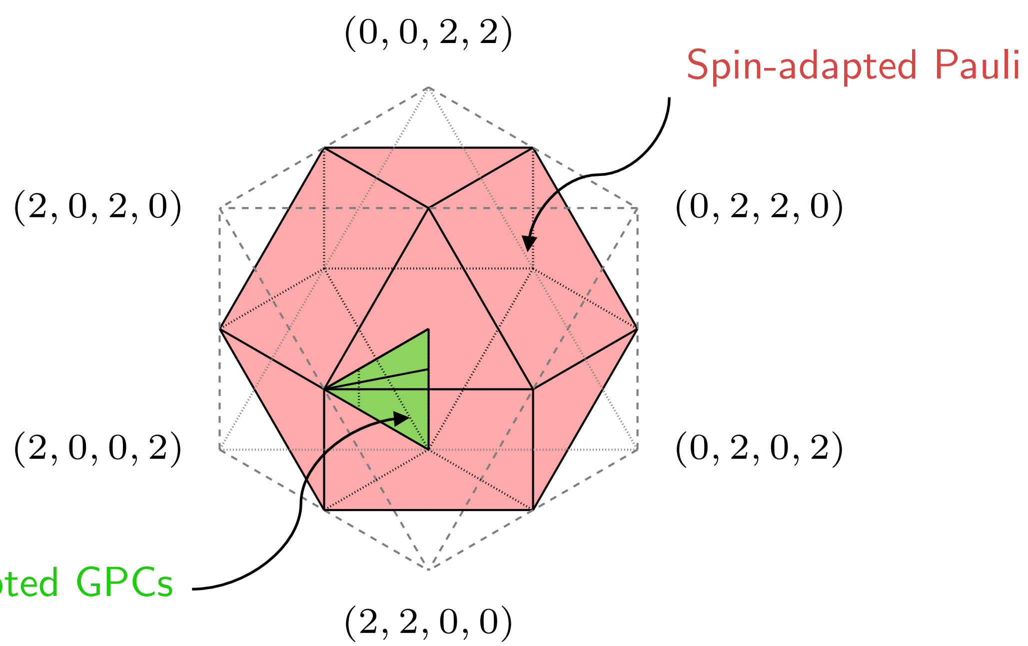

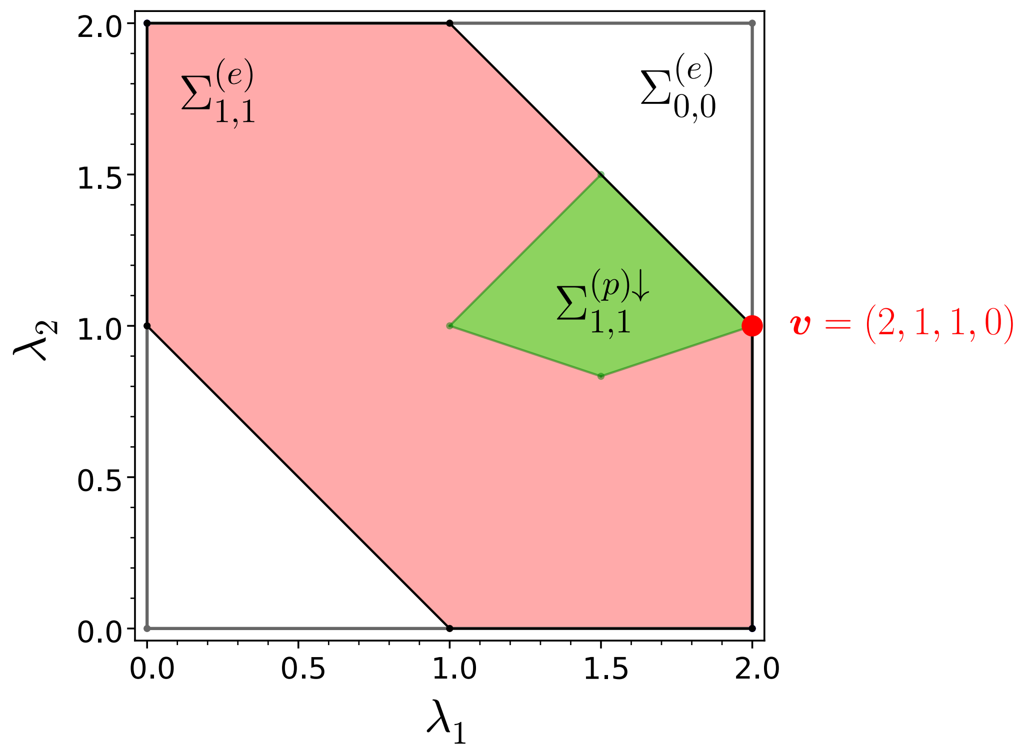

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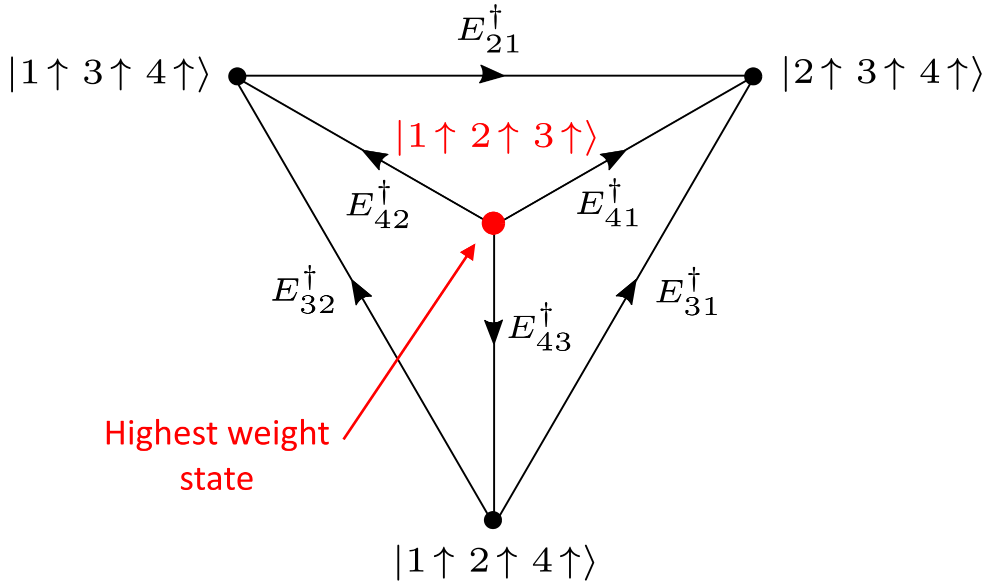
Spin-adapted ensemble N-representability problem

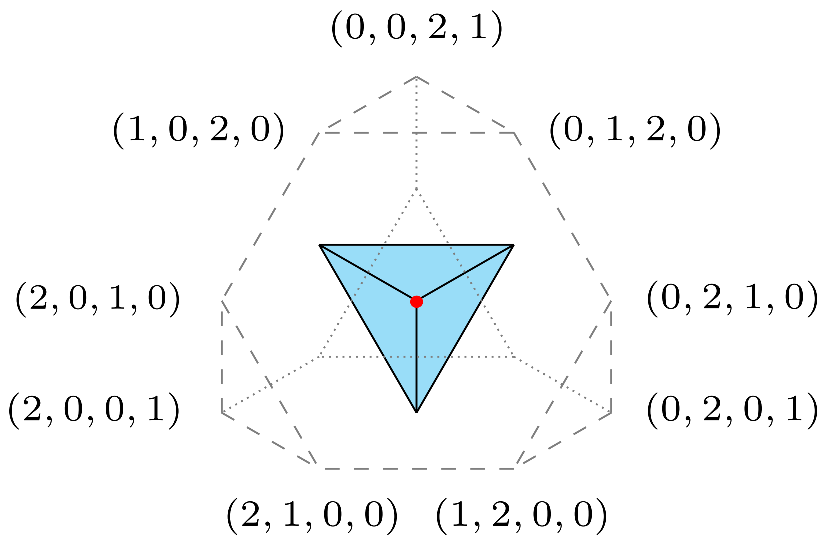
Generators of $U(d)$: $E_{ij} = f^\dagger_{j\uparrow} f_{i\uparrow} + f^\dagger_{j\downarrow} f_{i\downarrow} \implies$ Construct a Verma basis of $\mathcal{H}_N^{(S,M)}$

Example: $N = 3,\ S = M = \dfrac{3}{2}$

State hierarchy:



Spectral polytope:



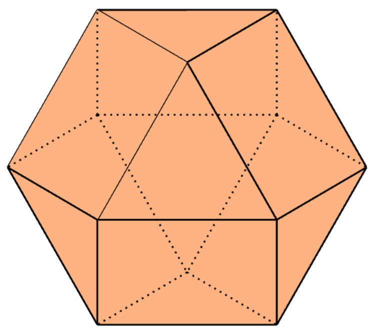
[D.-N. Verma, Bull, Amer. Math Soc. **74**, 160-166 (1968), S. Li, R.V. Moody, M. Nicolescu, J. Patera, J. Math. Phys. **27**, 668-677 (1986)]

Spin-adapted Pauli constraints

Orbital natural occupation numbers: $\boldsymbol{\lambda}^{\downarrow} = \mathrm{spec}^{\downarrow}(\gamma_l)$

$$\boxed{\lambda_1^{\downarrow} \leq 2\,, \quad \sum_{i=1}^{(N-2S)/2} \lambda_i^{\downarrow} \leq N - 2S\,, \quad \sum_{i=1}^{(N-2S)/2+1} \lambda_i^{\downarrow} \leq N - 2S + 1\,, \quad \sum_{i=1}^{(N+2S)/2} \lambda_i^{\downarrow} \leq N}$$

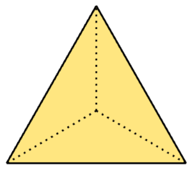
Example: $N = 4 = d = 4, S = 1$



Minkowski sum

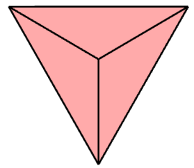
=

$N_{\downarrow} = 1$

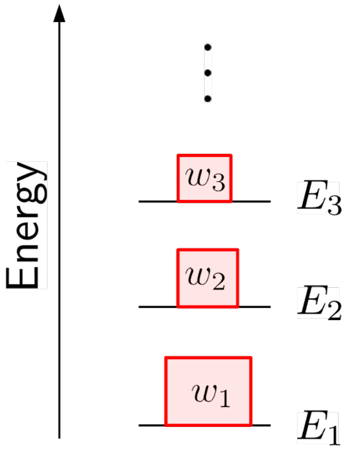


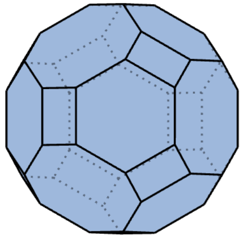
+

$N_{\uparrow} = 3$



III) Excited States



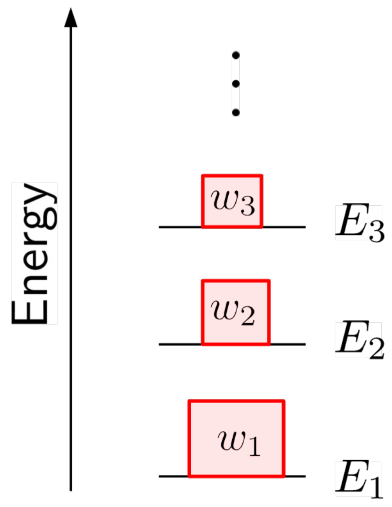


Key Concepts

1) Ensemble variational principle:

$$\boxed{E_{\boldsymbol{w}} = \sum_j w_j E_j = \min_{\Gamma \in \mathcal{E}^N(\boldsymbol{w})} \mathrm{Tr}_N[\Gamma H], \quad \Gamma \in \mathcal{E}^N(\boldsymbol{w})}$$

Set of N-particle states with fixed spectrum $\boldsymbol{w}$



[E.K.U. Gross, L.N. Oliveira, W. Kohn, Phys. Rev. A **37**, 2805 (1988)]

Key Concepts

2) Moment map and natural variable:

$$\boxed{\mu : \mathcal{E}^N(\boldsymbol{w}) \to \mathfrak{u}^*(2d)}$$

unitary
orbit

Convexity theorems:

$$\boxed{\Sigma^{\downarrow}(\boldsymbol{w}) := \mathcal{E}^1_N(\boldsymbol{w}) \cap \mathfrak{t}^*_+}$$

[B. Kostant, Ann. Sci. École Norm. Sup. **6**, 413-455 (1974),
M.F. Atiyah, Bull. London Math. Soc. **14**, 1-15 (1982),
V. Guillemin, S. Sternberg, Invent. Math. **67**, 491-513 (1982)]

Set of 1RDMs γ compatible
with an N-fermion state
$\Gamma \in \mathcal{E}^N(\boldsymbol{w})$

$\Longrightarrow$ Set of admissible (decreasingly ordered) natural occupation number vectors
forms a convex polytope

[F. Castillo, J.-P. Labbé, J. Liebert, A. Padrol, E. Philippe, C. Schilling, Ann. Henri, Poincaré **24**, 2241-2321 (2023)]

Key Concepts

2) Constrained search:

Ensemble variational principle

$$E_{\boldsymbol{w}} = \sum_j w_j E_j = \min_{\Gamma \in \mathcal{E}^N(\boldsymbol{w})} \mathrm{Tr}_N[\Gamma H]$$

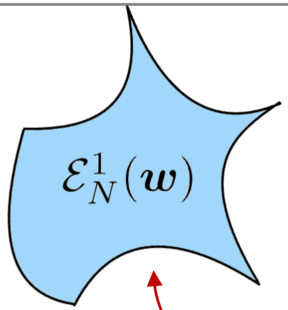
Moment map/pair of dual variables

$$\mu : \mathcal{E}^N(\boldsymbol{w}) \to i\mathfrak{u}^*(2d)$$
$$h \leftrightarrow \gamma$$

$$\Longrightarrow \quad \boxed{E_{\boldsymbol{w}} \equiv \min_{\gamma \in \mathcal{E}^1_N(\boldsymbol{w})} \left[\mathrm{Tr}_1[h\gamma] + \mathcal{F}_{\boldsymbol{w}}(\gamma)\right]}$$

Domain?

$\mathcal{F}_{\boldsymbol{w}}(\gamma)$:



$\mapsto \mathbb{R}$

Too complicated!

Key Concepts

2) Constrained search:

Ensemble variational principle

$$E_{\boldsymbol{w}} = \sum_j w_j E_j = \min_{\Gamma \in \mathcal{E}^N(\boldsymbol{w})} \mathrm{Tr}_N[\Gamma H]$$

Moment map/pair of dual variables

$$\mu : \mathcal{E}^N(\boldsymbol{w}) \to i\mathfrak{u}^*(2d)$$
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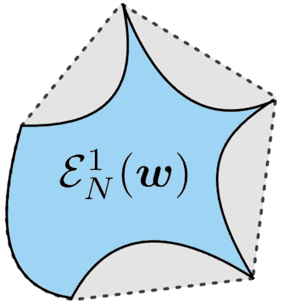
Domain?

$\mathcal{F}_{\boldsymbol{w}}(\gamma)$:

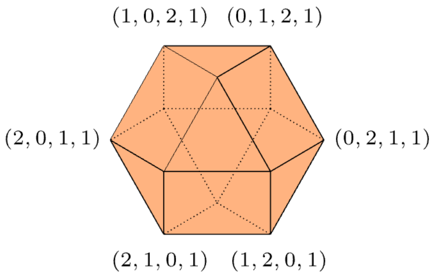


$\mapsto \mathbb{R}$

$\Longrightarrow$ **Exact convex relaxation!**

Spin-adapted Constraints

Spin-adapted Pauli constraints



+

$r=3$:

$r=2$:

$$2\sum_{i=1}^{(N-2S)/2}\lambda_i^{\downarrow}+\sum_{i=(N-2S)/2+1}^{(N+2S)/2}\lambda_i^{\downarrow}\leq 2(N-S)-1+w_1\,,$$

$$3\sum_{i=1}^{(N-2S)/2-1}\lambda_i^{\downarrow}+2\sum_{i=(N-2S)/2}^{(N-2S)/2+1}\lambda_i^{\downarrow}+\sum_{i=(N-2S)/2+2}^{(N+2S)/2}\lambda_i^{\downarrow}\leq 3N-4S-2+w_1+w_2\,,$$

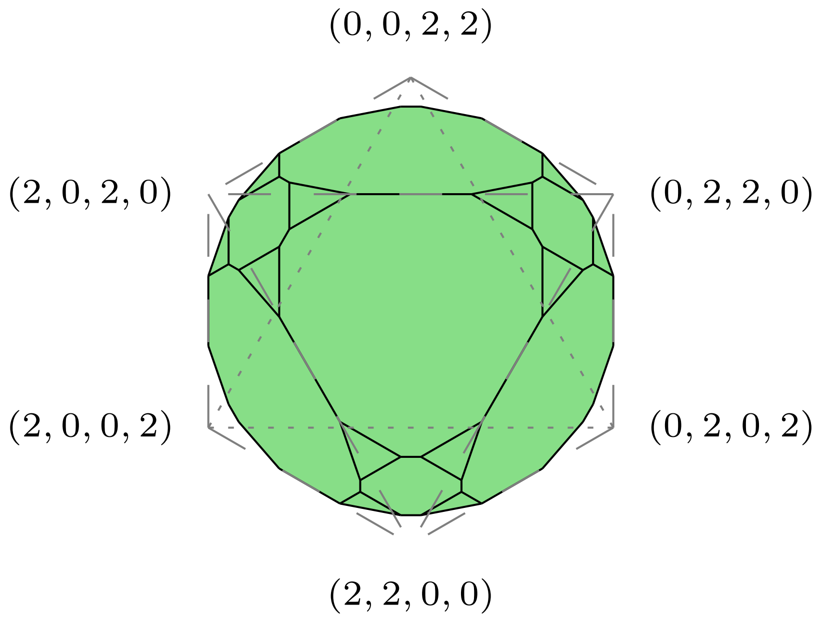
$$3\sum_{i=1}^{(N-2S)/2}\lambda_i^{\downarrow}+2\sum_{i=(N-2S)/2+1}^{(N+2S)/2-1}\lambda_i^{\downarrow}+\sum_{i=(N+2S)/2}^{(N+2S)/2+1}\lambda_i^{\downarrow}\leq 3N-2S-2+w_1+w_2$$

[J. Liebert, F. Castillo, J.-P. Labbé, T. Maciazek, C. Schilling, Quantum **9**, 1921 (2025)]

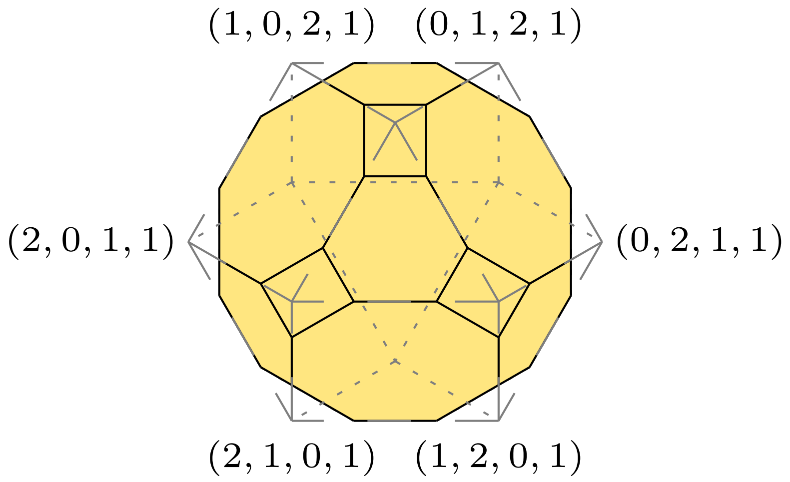
$\Downarrow$

Spin-dependent $\boldsymbol{w}$-ensemble orbital one-body N-representability constraints

Singlet



Triplet



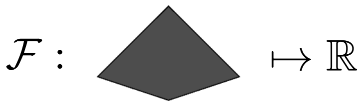
[J. Liebert, F. Castillo, J.-P. Labbé, T. Maciazek, C. Schilling, Quantum **9**, 1921 (2025)]

Summary and Conclusions

- Moment maps appear naturally in functional theories

  $\Rightarrow$ Choice of natural variable determined by dual pairing

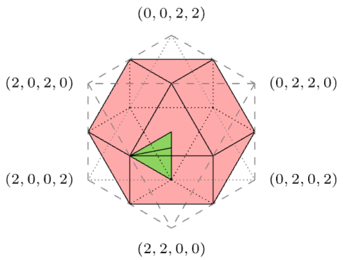
  $\Rightarrow$ Domain of functional often a moment polytope



- Spin-adapted N-representability problem and functional theory

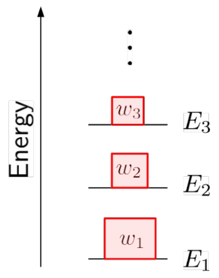
  $\Rightarrow$ Natural variable in RDMFT simplifies

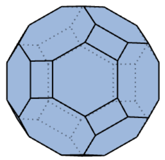
  $\Rightarrow$ Spin-adapted Pauli constraints



- Foundation of ensemble RDMFT for excited states

  $\Rightarrow$ Generalization of Pauli's exclusion principle for mixed states



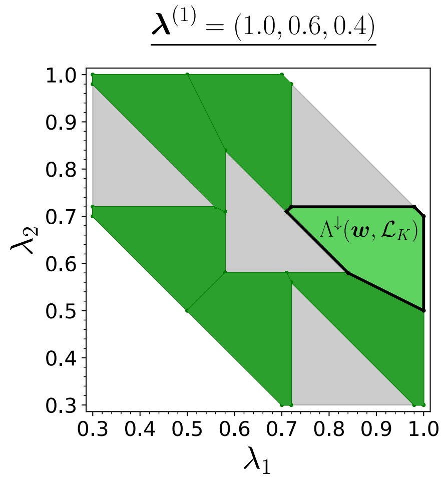


Outlook

- Refinement based on partial information

? ***"How does the set of admissible 1RDMs change if we assume partial knowledge about some of the ensemble members?"***

[J. Liebert, A. Schouten, I. Avdic, C. Schilling, D. Mazziotti, New J. Phys. **27**, 124511 (2025)]

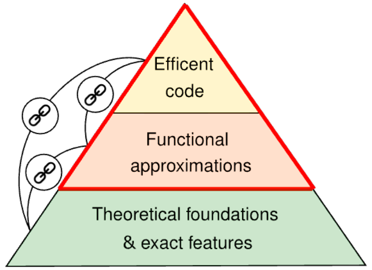


- Solution to spin-adapted N-representability problem for *full* 1RDM

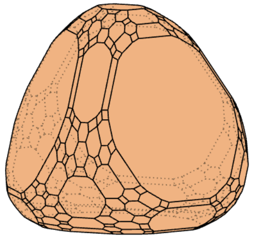
$\Rightarrow$ Constraints on both natural occupation numbers and natural orbitals

$\Rightarrow$ Complexity class?

- Development and testing of functional approximations



Thank you!



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